

The Hybrid Semantic Lake Architecture

Abstract

This document specifies the Hybrid Semantic Lake Architecture. Building upon the normalized artifacts of the Faceted Plain-Text Lake (RFC 0001), this specification introduces Vector Embeddings and Metadata-Filtered Retrieval-Augmented Generation (RAG). By combining deterministic faceting with associative spatial topology, the system achieves highly accurate, context-aware semantic retrieval.

1. Introduction

While the Faceted Plain-Text Lake solves the physical shelf limitation, it still relies on exact keyword or deterministic metadata matching. The Hybrid Semantic Lake introduces "cue-dependent navigation" similar to human cognitive architectures (e.g., the Method of Loci). It maps unstructured information into a spatial mathematical topology, allowing retrieval based on semantic proximity rather than literal terminology.

2. Vector Embedding Integration

In this phase, all Markdown documents from RFC 0001 are processed by an embedding model. The text is translated into high-dimensional numerical vectors and stored within a Vector Database.

This creates an associative map where concepts sharing semantic similarities are located near each other in the vector space, discarding the need for a rigid top-down directory tree entirely.

3. Metadata-Filtered RAG

Pure vector search is prone to hallucination due to word sense ambiguity across disparate disciplines. The defining innovation of the Hybrid Semantic Lake is the marriage of RFC 0001's YAML Frontmatter with the Vector Database.

During the ingestion phase, the facets (Industry, Domain, Process) are attached to the vector payload as filterable metadata properties.

4. The Query Execution Flow

When a user or application submits a natural language query, the architecture **MUST** process the request using a two-stage deterministic-semantic funnel:

1. **Intent Routing & Deterministic Filtering:** The system analyzes the query to extract structured parameters. It applies a hard deterministic filter to the Vector Database (e.g., `PRE-FILTER: domain == 'Engineering'`).
2. **Associative Semantic Search:** The user's query is converted into a vector and plotted into the pre-filtered sector of the database. The system retrieves the k-nearest neighbors based on mathematical proximity (Cosine Similarity or Euclidean Distance).
3. **Generative Synthesis:** The retrieved Markdown chunks are provided to a Large Language Model to synthesize a coherent, grounded response (RAG).

5. Knowledge Graph Enhancement (Optional)

Implementations **MAY** introduce a Knowledge Graph layer atop the vector store. This explicitly defines the edges (relationships) between the nodes (documents), providing structured traversal paths that complement the fuzzy logic of spatial vector search.